

CS 105: Department Introductory Course on Discrete Structures

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Lecture 10 – Basic structures: relations: posets

Recap and moving on...

Week 01, 02, 03: Proofs and Reasoning

- ▶ Propositions, Predicates, Quantifiers
- ▶ Theorems and Types of Proofs
- ▶ Induction and variants

Week 03, 04, 05: Basic Mathematical Structures

- ▶ Finite and infinite sets.
- ▶ Using functions to compare sets: focus on bijections.
- ▶ Countable, countably infinite and uncountable sets.
- ▶ Cantor's diagonalization argument (A new powerful proof technique!).

Cantor's Proof

For any set S , there is no bijection between S and $\mathcal{P}(S)$.

1. Suppose there is such a bijection, say f .
2. Consider $S' = \{x \in S \mid x \notin f(x)\} \in \mathcal{P}(S)$
3. But f is a surjection, so there exists $y \in S$ such that $f(y) = S'$.
4. So is $y \in S'$?
 - 4.1 If $y \in S'$, then $y \notin f(y) = S'$ a contradiction.
 - 4.2 If $y \notin S'$, then $y \notin f(y)$, so $y \in S'$, a contradiction!
5. Hence both cases lead to contradiction, so our premise must be false, i.e., there is no such bijection.

Last class: Relations

Basic Mathematical Structures – Relations

- ▶ Equivalence relations: reflexive, symmetric, transitive
- ▶ Equivalence relations and partitions of a set
- ▶ Equivalence classes
- ▶ Application: Using equivalence relations to define new mathematical objects

Moving on another special relation: Partial Orders

Consider $\{(a, b) \mid a, b \in \mathbb{Z}, a \leq b\}$. This is reflexive, transitive but not symmetric. In fact, it is quite different!

Anti-symmetric

A relation R on S is **anti-symmetric** if for all $a, b \in S$ aRb and bRa implies $a = b$.

Examples:

- ▶ $R_1(\mathbb{Z}) = \{(a, b) \mid a, b \in \mathbb{Z}, a \leq b\}$.
- ▶ $R_2(\mathcal{P}(S)) = \{(A, B) \mid A, B \in \mathcal{P}(S), A \subseteq B\}$.

Definition

A **partial order** is a relation which is **reflexive**, **transitive** and **anti-symmetric**.

Partial orders and equivalences relations

- ▶ **Reflexive:** $\forall a \in S, aRa$.
- ▶ **Symmetric:** $\forall a, b \in S, aRb$ implies bRa .
- ▶ **Anti-symmetric:** $\forall a, b \in S, aRb, bRa$ implies $a = b$.
- ▶ **Transitive:** $\forall a, b, c \in S, aRb, bRc$ implies aRc .

	Refl.	Anti-Sym	Trans.	PO
$\{(a, b) \mid a, b \in \mathbb{Z}, a \leq b\}$	✓	✓	✓	✓
$\{(A, B) \mid A, B \in \mathcal{P}(S), A \subseteq B\}$	✓	✓	✓	✓
$\{(a, b) \mid a, b \in \mathbb{Z}, a < b\}$				
$\{(a, b) \mid a, b \in \mathbb{Z}^+, a \mid b\}$				
$\{((a, b), (c, d)) \mid (a, b), (c, d) \in \mathbb{Z}^+ \times \mathbb{Z}^+, a < c \text{ or } (a = c, b \leq d)\}$				

Partial orders

	Refl.	Anti-Sym	Trans.	PO	TO
$\{(a, b) \mid a, b \in \mathbb{Z}, a \leq b\}$	✓	✓	✓	✓	✓
$\{(A, B) \mid A, B \in \mathcal{P}(S), A \subseteq B\}$	✓	✓	✓	✓	×
$\{(a, b) \mid a, b \in \mathbb{Z}, a < b\}$					
$\{(a, b) \mid a, b \in \mathbb{Z}^+, a \mid b\}$					
$\{((a, b), (c, d)) \mid (a, b), (c, d) \in \mathbb{Z}^+ \times \mathbb{Z}^+, a < c \text{ or } (a = c, b \leq d)\}$					

- ▶ We use $\preceq$ to denote partial orders and write $a \preceq b$ instead of aRb .
- ▶ Why is it called “partial” order? Because, not all pairs of elements are “comparable” (i.e., related by $\preceq$).
- ▶ A **total order** is a partial order $\preceq$ on S in which every pair of elements is comparable
- ▶ Qn: Can a relation be symmetric and anti-symmetric?
- ▶ Qn: Can a relation be neither symmetric nor anti-symmetric?

Partially ordered sets (Posets)

Definition

A set S together with a partial order $\preceq$ on S , is called a **partially-ordered set** or **poset**, denoted $(S, \preceq)$.

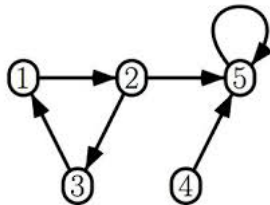
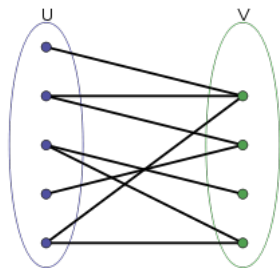
Examples

- ▶ $(\mathbb{Z}, \leq)$: integers with the usual less than or equal to relation.
- ▶ $(\mathcal{P}(S), \subseteq)$: powerset of any set with the subset relation.
- ▶ $(\mathbb{Z}^+, |)$: positive integers with divisibility relation.

Graphical representation of relations: posets

Recall: any relation on a set can be represented as a **graph** with

- ▶ nodes as elements of the set and
- ▶ directed edges between them indicating the ordered pairs that are related.



- ▶ Did these come from posets?
- ▶ Do graphs defined by posets have any “special” properties?

Graphical representation of relations: posets

- Let $S = \{1, 2, 3\}$. Recall the poset $(\mathcal{P}(S), \subseteq)$.

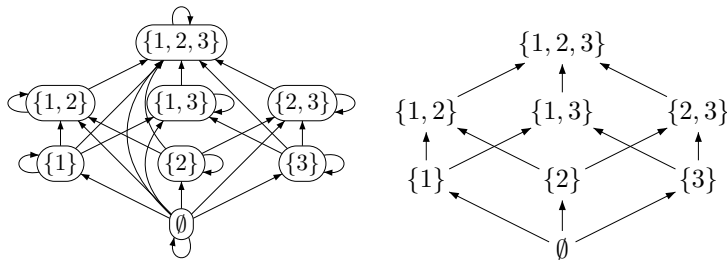


Figure: Graph of a poset and its **Hasse diagram**

- What is “special” about these graphs?
- **Graphs of posets are “acyclic” (except for self-loops).**
- Starting from a node and following the directed edges (except self-loops), one can’t come back to the same node.
- The **reflexive transitive closure** of Hasse diagram gives back graph of the poset.

Chains and Anti-chains

Definition

Let $(S, \preceq)$ be a poset. A subset $B \subseteq S$ is called

- ▶ a **chain** if every pair of elements in B is related by $\preceq$.
- ▶ That is, $\forall a, b \in B$, we have $a \preceq b$ or $b \preceq a$ (or both).
- ▶ Thus, $\preceq$ is a total order on B .

Definition

Let $(S, \preceq)$ be a poset. A subset $A \subseteq S$ is called

- ▶ an **anti-chain** if no two distinct elements of A are related to each other under $\preceq$.
- ▶ That is, $\forall a, b \in A, a \neq b$, we have neither $a \preceq b$ nor $b \preceq a$.

Chains and Anti-chains: examples

- Let $S = \{1, 2, 3\}$.

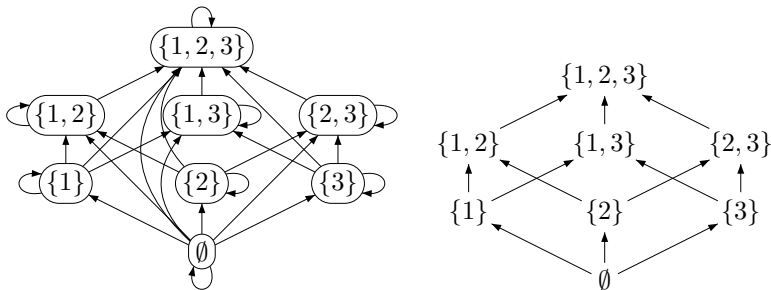


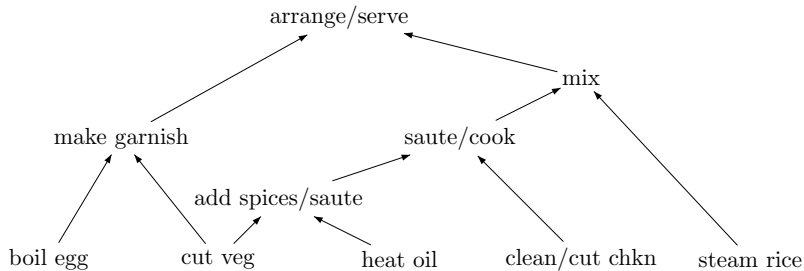
Figure: Graph of poset $(\mathcal{P}(S), \subseteq)$ and its Hasse diagram

- What are the chains in this poset?
- What are the anti-chains in this poset?

Examples and applications

A task scheduling example

Let us represent a recipe for making Chicken Biryani as a poset!



- ▶ Clearly, this shows the **dependencies**.
- ▶ But when you cook you need a total order, right?
- ▶ Further, this total order must be consistent with the po.
- ▶ This is called a **linearization** or a **topological sorting**.

Topological sorting

Definition

A **topological sort** or a **linearization** of a poset $(S, \preceq)$ is a poset $(S, \preceq_t)$ with a total order $\preceq_t$ such that $x \preceq y$ implies $x \preceq_t y$.

Theorem

Every finite poset has a topological sort.

Proof: (H.W)

- ▶ First prove the following lemma:
 - ▶ Every finite non-empty poset has at least one minimal element (x is minimal if $\nexists y, y \preceq x$).
- ▶ Then, construct the chain to complete the proof.